

SIMATS ENGINEERING

ASSESSMENT TOOL 1: AI Model Design Exercise

COURSE NAME: **ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01**
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A AI Model Design Exercise

Code of Conduct:

I SHAM S 192325076 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Dataset Selection and Data Understanding	10%	
3. Data Preprocessing and Feature Engineering	10%	
4. AI Technique Selection and Justification	10%	
5. Model Design / Architecture	10%	
6. Algorithm / Pseudocode Development	5%	
7. Model Implementation and GitHub Repository	15%	
8. Experimental Design and Results	10%	
9. Performance Evaluation and Metrics	10%	

10. Result Analysis and Interpretation	5%	
11. Limitations and Future Enhancements	5%	
12. Documentation and Technical Presentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Scenario / Problem Statement	AI Model / Technique to be Developed	Expected Development Outcome
1	Smart Loan Approval: A bank wants to automatically determine whether a customer is eligible for a loan using income, credit score, employment status, loan amount, and repayment history.	Decision Tree	Design and implement a decision-tree-based AI model and determine the most suitable attribute-selection measure such as Information Gain, Gain Ratio, or Gini Index.
2	Student Performance Prediction: A university wants to predict whether a student is likely to pass or fail a course using attendance, internal marks, assignment performance, study hours, and previous academic performance.	Inductive Learning	Construct a predictive model from training examples and derive generalized decision rules from the learned patterns.
3	Disease Risk Prediction: A healthcare system needs to classify patients into low-risk and high-risk categories based on clinical measurements.	Statistical Learning	Design a statistical learning model, train it using patient data, and evaluate its generalization ability on unseen data.

4	Image Recognition System: An organization wants to automatically classify images into categories such as vehicles, animals, and buildings.	Neural Network	Design and implement a neural-network classifier by specifying the network architecture, activation functions, loss function, optimizer, and training procedure.
5	Handwritten Character Recognition: A system must recognize handwritten characters from scanned images.	Multilayer Neural Network	Develop a multilayer neural network and demonstrate how backpropagation updates weights to minimize classification error.
6	Non-Linear Classification: A dataset contains two classes that cannot be separated using a straight-line decision boundary.	SVM with Kernel Method	Design an SVM-based classifier using an appropriate kernel function and justify the selected kernel based on the characteristics of the dataset.
7	Intelligent Recommendation Agent: An online platform wants an agent that learns which recommendations to provide based on user responses.	Reinforcement Learning	Design an RL model by defining the states, actions, rewards, policy, and learning strategy, and demonstrate how the agent learns from user feedback.
8	Autonomous Game Agent: An AI agent must learn to play a game where successful actions receive positive rewards and unsuccessful actions receive penalties.	Reinforcement Learning	Develop an RL-based game agent and demonstrate how its policy and performance improve through repeated interaction with the environment.
9	Explainable Diagnosis System: A medical AI system predicts a disease but must also provide an explanation for its prediction.	Explanation-Based Learning	Design an explanation-based learning approach using domain knowledge and training examples to generate an understandable justification for the prediction.

10	AI Model Selection Challenge: A company provides a dataset for predicting customer churn.	Comparative AI Model Design	Design and compare at least two learning approaches from decision trees, statistical learning, neural networks, or kernel methods. Select the best model based on accuracy, interpretability, computational requirements, and generalization ability.
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Sure. Here is the **complete final content** for your PDF/report, using the **actual results you obtained**.

SMART LOAN APPROVAL PREDICTION USING DECISION TREE CLASSIFICATION

Course: Artificial Intelligence and Expert Systems

Course Code: MLA01

Assessment: Tool 1 – AI Model Design Exercise

Topic: Smart Loan Approval

Student: Sham S.

GitHub: Sham-2005/AI_project/Assignment_tool/CO4

1. Problem Statement

Loan approval is an important decision-making process in the banking and financial sector. A loan application is generally evaluated using several customer-related factors such as income, credit score, employment status, loan amount, and previous repayment behaviour.

The objective of this project is to develop an Artificial Intelligence-based system that automatically predicts whether a customer is eligible for a loan.

The proposed system uses a **Decision Tree Classification algorithm** to learn relationships between customer attributes and loan eligibility.

The system produces one of two outputs:

- **Eligible**
- **Not Eligible**

The project also compares two attribute-selection criteria:

- **Gini Index**
- **Information Gain using Entropy**

The purpose of this comparison is to determine which criterion provides better classification performance for the loan approval problem.

2. Objectives

The main objectives of the project are:

1. To develop an AI-based loan eligibility prediction system.
 2. To preprocess loan application data for machine learning.
 3. To identify important factors influencing loan eligibility.
 4. To implement a Decision Tree classification algorithm.
 5. To compare Gini Index and Information Gain.
 6. To evaluate the model using suitable classification metrics.
 7. To visualize the Decision Tree structure.
 8. To analyse feature importance.
 9. To test the system with a new customer.
 10. To provide a reproducible implementation through GitHub.
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3. Dataset Description

A **synthetic demonstration dataset** was generated using Python because an external loan dataset was not supplied during implementation.

The dataset contains **1,000 loan application records** and **6 columns**.

Dataset Features

Feature	Description	Type
Income	Customer's income	Numerical
Credit Score	Customer's creditworthiness score	Numerical
Employment Status	Employment category	Categorical
Loan Amount	Amount requested by the customer	Numerical
Repayment History	Previous repayment behaviour	Categorical
Loan Status	Loan eligibility	Target

Target Distribution

Loan Status Number of Records

Eligible	487
Not Eligible	513
Total	1,000

The target classes are reasonably balanced, with 487 Eligible and 513 Not Eligible records.

4. Data Preprocessing

Data preprocessing was performed before training the Decision Tree model.

4.1 Duplicate Removal

Duplicate records were identified and removed to prevent repeated observations from affecting model training.

4.2 Missing Value Handling

The dataset was checked for missing values.

The final generated dataset contained:

0 missing values in all six columns.

For general preprocessing, numerical missing values can be replaced using the median, while categorical missing values can be replaced using the mode.

4.3 Categorical Encoding

The following categorical features were converted into numerical values:

- Employment Status
- Repayment History

Label Encoding was used for categorical variables.

4.4 Feature and Target Separation

The five input features were separated from the target variable.

Input features:

- Income
- Credit Score
- Employment Status

- Loan Amount
- Repayment History

Target:

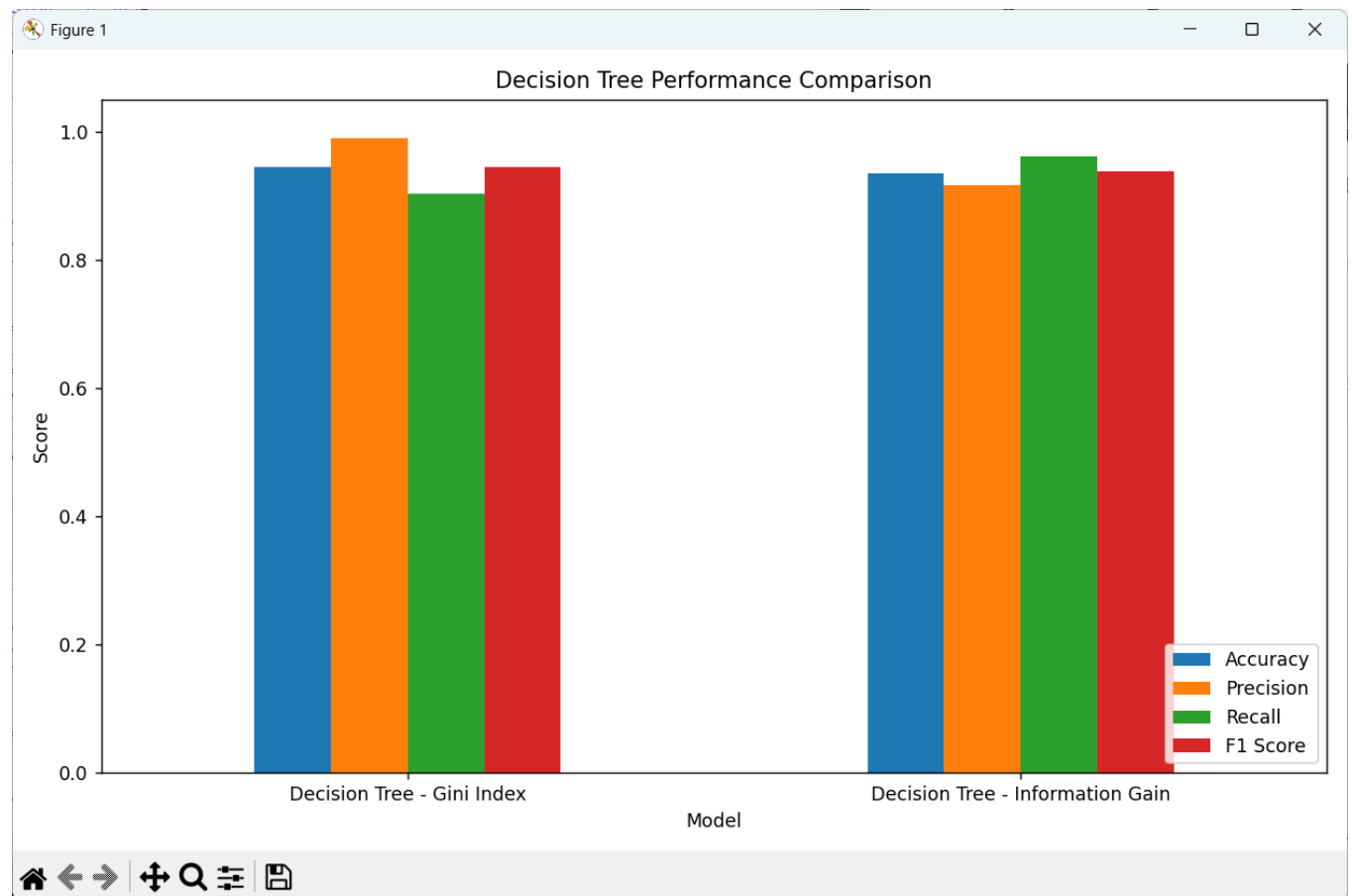
- Loan Status

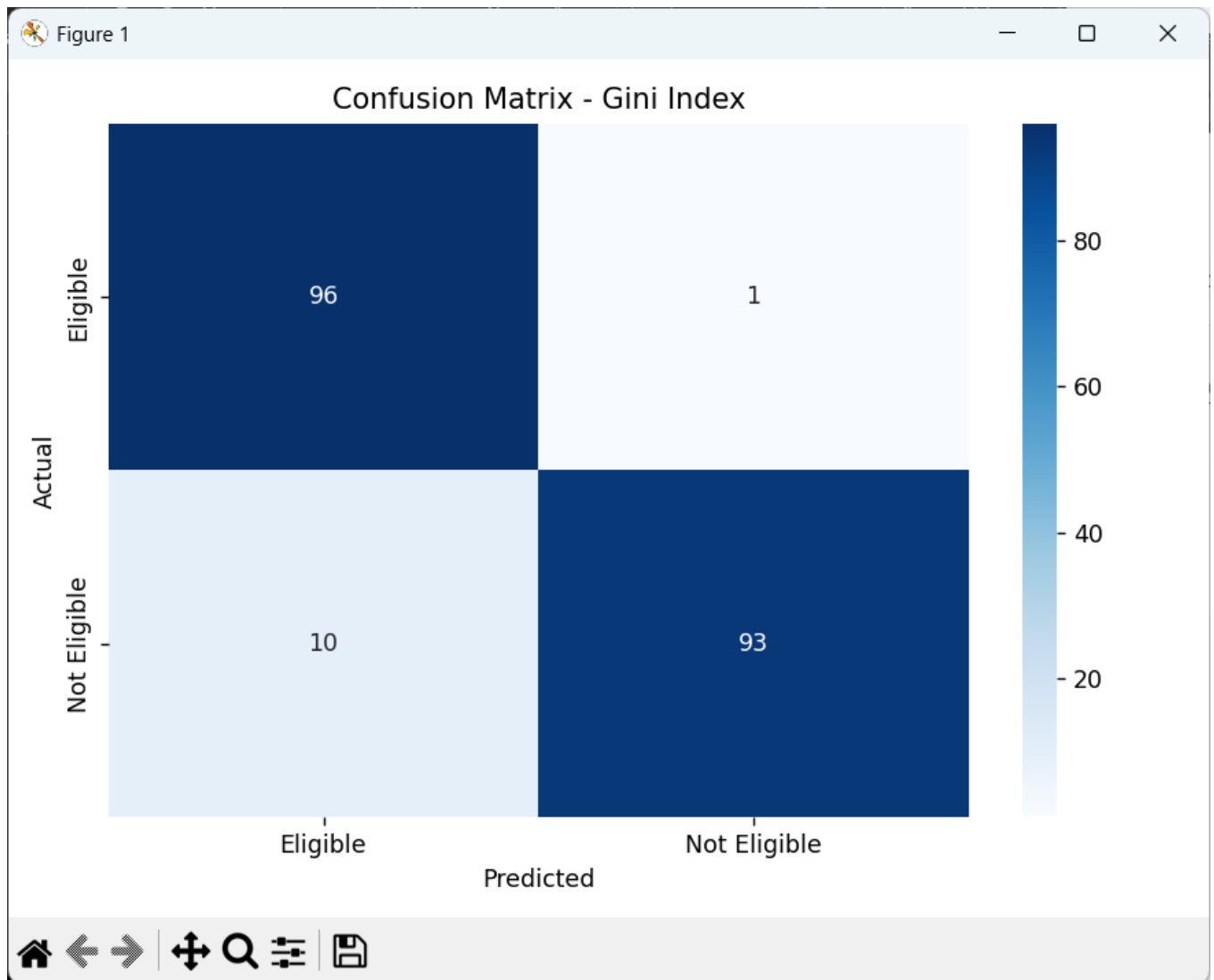
4.5 Train-Test Split

The dataset was divided into:

- **80% training data = 800 records**
- **20% testing data = 200 records**

A fixed random state of 42 was used to ensure reproducibility.





5. Selected AI Technique and Justification

Decision Tree Classification

A **Decision Tree Classifier** was selected because the problem is a binary classification problem.

The Decision Tree learns a series of decision rules from the input features and uses them to classify customers.

Reasons for selecting Decision Tree

1. Suitable for classification problems.
2. Works well with structured/tabular data.
3. Easy to understand and interpret.
4. Decision rules can be visualized.

5. Feature importance can be extracted.
6. Different splitting criteria can be compared.
7. Requires relatively little preprocessing.
8. Suitable for demonstrating AI decision-making.

The two splitting criteria used in this project are:

Gini Index

Gini Index measures the impurity of the data at a particular node.

Lower impurity indicates a better split.

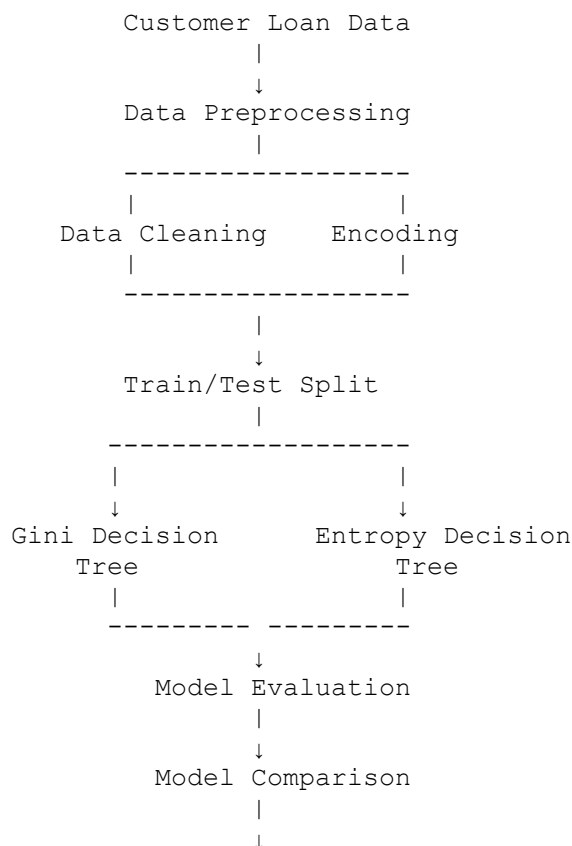
Information Gain

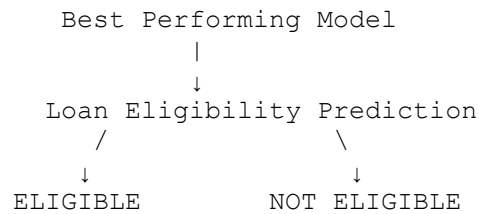
Information Gain uses **Entropy** to measure the reduction in uncertainty after a split.

Higher information gain indicates a better split.

6. Model Design / Architecture

The proposed system follows the following architecture:





Model Parameters

The Decision Tree was configured with:

Criterion : Gini / Entropy
Maximum Depth : 5
Minimum Samples : 10
Random State : 42

7. Algorithm / Pseudocode

START

1. Load the loan dataset.
2. Check for missing values and duplicate records.
3. Remove duplicate records.
4. Handle missing values.
5. Encode categorical variables.
6. Separate input features X and target Y.
7. Split the dataset:
 - 80% Training
 - 20% Testing
8. Create Decision Tree using Gini Index.
9. Train the Gini model.
10. Predict the testing data.
11. Calculate:
 - Accuracy
 - Precision
 - Recall
 - F1-score
12. Create Decision Tree using Entropy.
13. Train the Entropy model.
14. Predict the testing data.
15. Calculate evaluation metrics.

16. Compare both models.
 17. Select the better-performing model.
 18. Generate confusion matrix.
 19. Calculate feature importance.
 20. Predict loan eligibility for a new customer.
- END
-

8. Implementation

The project was implemented using **Python**.

Technologies Used

- Python
- Pandas
- NumPy
- Scikit-learn
- Matplotlib
- Seaborn

Main Implementation Steps

1. Dataset loading/generation
2. Data inspection
3. Missing-value checking
4. Data preprocessing
5. Categorical encoding
6. Train-test splitting
7. Gini Decision Tree training
8. Entropy Decision Tree training
9. Model prediction
10. Performance evaluation
11. Model comparison
12. Confusion matrix generation
13. Feature-importance analysis
14. Decision Tree visualization
15. New customer prediction

Generated Output Files

loan_approval.csv
loan_predictions.csv
model_comparison.csv
model_comparison.png

confusion_matrix.png
feature_importance.csv
feature_importance.png
decision_tree.png

9. Experimental Results

Two Decision Tree models were developed and compared.

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score
Gini Index	94.50%	98.94%	90.29%	94.42%
Information Gain	93.50%	91.67%	96.12%	93.84%

Best Model

The **Gini Index Decision Tree** was selected as the best overall model.

It achieved:

- **Accuracy: 94.50%**
 - **Precision: 98.94%**
 - **Recall: 90.29%**
 - **F1-Score: 94.42%**
-

10. Performance Evaluation

The models were evaluated using four major metrics.

Accuracy

Accuracy measures the percentage of correctly classified loan applications.

For the Gini model:

Accuracy = 94.50%

This means the model correctly classified approximately 94.5% of the test samples.

Precision measures how many of the customers predicted as a particular positive class were correctly classified.

For the Gini model:

Precision = 98.94%

This indicates very few false-positive predictions for the evaluated positive class.

Recall

Recall measures how many of the actual positive cases were correctly identified.

For the Gini model:

Recall = 90.29%

The Information Gain model achieved a higher recall of **96.12%**.

F1-Score

F1-score provides a balance between precision and recall.

For the Gini model:

F1-Score = 94.42%

For the Information Gain model:

F1-Score = 93.84%

Therefore, the Gini model achieved a slightly better balance between precision and recall.

11. Result Analysis and Interpretation

The experimental results show that both Decision Tree models performed well on the test dataset.

The **Gini-based Decision Tree achieved 94.50% accuracy**, which was higher than the **93.50% accuracy** achieved by the Information Gain model.

The Gini model also achieved a higher precision of **98.94%**, compared with **91.67%** for Information Gain.

However, the Information Gain model achieved a higher recall of **96.12%**, compared with **90.29%** for the Gini model.

The F1-score of the Gini model was **94.42%**, slightly higher than the Information Gain model's **93.84%**.

Therefore, considering accuracy, precision and F1-score together, the **Gini Index Decision Tree** was selected as the final model.

Feature Importance Analysis

The feature importance generated by the selected Gini model is:

Rank	Feature	Importance
1	Repayment History	36.48%
2	Credit Score	35.34%
3	Income	16.52%
4	Loan Amount	5.92%
5	Employment Status	5.74%

Interpretation

Repayment History was the most important feature, with an importance of **36.48%**.

Credit Score was the second most important feature, with **35.34%**.

Together, these two features account for approximately:

$$36.48 + 35.34 = 71.82\%$$

of the total feature-importance score.

This indicates that repayment behaviour and creditworthiness are the most influential attributes used by the trained model in this experiment.

12. Sample Customer Prediction

The final model was tested using the following sample customer:

Attribute	Value
Income	₹75,000
Credit Score	720
Employment Status	Employed
Loan Amount	₹4,00,000
Repayment History	Good

Prediction

Loan Status: ELIGIBLE

The model classified the sample customer as **Eligible**.

The prediction probability returned by the Decision Tree was:

```
[[1.0, 0.0]]
```

This represents the model's probability output for this particular sample and should not be interpreted as absolute real-world certainty.

13. Limitations

The project has the following limitations:

1. The current experiment uses a synthetic demonstration dataset.
 2. The dataset contains only 1,000 records.
 3. Only five input features are considered.
 4. Real-world loan decisions may involve many additional financial factors.
 5. Synthetic data may contain simplified relationships that do not fully represent real-world behaviour.
 6. Decision Trees can overfit if their complexity is not controlled.
 7. The quality of predictions depends heavily on the quality and representativeness of the dataset.
 8. Historical or generated data may contain bias.
 9. The model should not completely replace human financial assessment.
 10. The model requires further validation before real-world deployment.
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14. Future Enhancements

The project can be enhanced in several ways.

1. Real-World Dataset

Replace the synthetic dataset with a verified real-world loan approval dataset.

2. Additional Features

Additional attributes such as:

- Age
- Existing debt
- Number of dependents
- Credit history length
- Previous defaults
- Monthly expenses

can be considered.

3. Hyperparameter Tuning

Parameters such as:

- Maximum depth
- Minimum samples split
- Minimum samples leaf

can be optimized.

4. Cross-Validation

K-fold cross-validation can be used to obtain a more reliable estimate of model performance.

5. Algorithm Comparison

The Decision Tree can be compared with:

- Random Forest
- Logistic Regression
- Support Vector Machine
- Gradient Boosting

6. Explainable AI

The system can provide an explanation for each prediction, such as:

"Eligible because of high credit score and good repayment history."

7. Web Application

A web interface can be developed where users enter their information and receive a loan eligibility prediction.

15. Conclusion

This project successfully demonstrates an AI-based **Smart Loan Approval Prediction System** using Decision Tree classification.

The system uses five important customer attributes:

- Income
- Credit Score
- Employment Status
- Loan Amount
- Repayment History

Two Decision Tree models were developed using **Gini Index** and **Information Gain**.

The Gini-based model achieved the best overall performance with:

94.50% Accuracy

98.94% Precision

90.29% Recall

94.42% F1-Score

Feature-importance analysis showed that **Repayment History** and **Credit Score** were the two most influential features, together accounting for approximately **71.82%** of the model's feature-importance score.

The project demonstrates the complete AI workflow from data preprocessing and model development to evaluation, visualization and prediction.

However, because the current experiment uses a synthetic dataset, a real-world dataset and additional validation would be required before applying the model to actual financial loan decisions.

16. GitHub Repository

Repository

GitHub: Sham-2005/AI_project

Project Path

Assignment_tool/CO4

Repository Contents

CO4/

```
|— smart_loan_approval.py
|— loan_approval.csv
|— loan_predictions.csv
|— model_comparison.csv
|— model_comparison.png
|— confusion_matrix.png
|— feature_importance.csv
|— feature_importance.png
|— decision_tree.png
```

File Description

File	Description
smart_loan_approval.py	Complete Python implementation
loan_approval.csv	Synthetic loan dataset
loan_predictions.csv	Model predictions
model_comparison.csv	Gini vs Information Gain results
model_comparison.png	Performance comparison graph
confusion_matrix.png	Confusion matrix
feature_importance.csv	Feature importance values
feature_importance.png	Feature importance graph
decision_tree.png	Decision Tree visualization

The repository provides the source code and generated experimental outputs required to reproduce the project.

References

1. SIMATS Engineering, **Assessment Tool 1 – AI Model Design Exercise**, Artificial Intelligence and Expert Systems, MLA01.
2. Scikit-learn documentation – Decision Tree Classifier.
3. Scikit-learn documentation – Classification Metrics.
4. Pandas documentation – Data Processing and Analysis.
5. NumPy documentation – Numerical Computing.
6. Matplotlib documentation – Data Visualization.
7. Seaborn documentation – Statistical Data Visualization.
8. Project source code and results – GitHub repository: Sham-2005/AI_project/Assignment_tool/CO4.